Detect Alternative Insect-Based Ingredients For Solubility Properties

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***Abstract*— Proteins play an important role in the food-processing industry. In this paper, we propose a protein property prediction tool for the case of solubility prediction of proteins by extracting its various features. In the proposed tool, a dataset consisting of a number of amino acids was used for protein properties prediction. Several models were made and ensemble learning was used to enhance the output.**

***Index Terms*—protein property prediction, ensemble learning, graph machine learning, neural networks, insect-based ingredient discovery, solubility prediction**

I. Introduction

The food-processing industry utilizes a variety of (artificial) additives to improve the preservation, taste, texture, and cosmetics of food-based products, among other reasons. Throughout the past decade, various studies have shown that the additives used within processed foods may contain potentially harmful proteins for human consumption [1] [2]. In recent years, there has been an increased interest in using better, safer, and more sustainable sources of food additives, ingredients, and other materials.

The manual process of performing said research is a time and resource consuming process. In an attempt to minimize the process, various institutes have spent researches on the prediction of protein-based elements and the creation of various databases. Similarly, this challenge will be addressed in this paper for the case of solubility prediction of proteins.

Previous research has mainly focused on predicting protein structures (Alphafold) [3] [4], protein alignments (PSI-BLAST) [5], secondary structures (SPIDER3 and ProteinUnet) [6] [7], protein alignments with respect to locations (HH-Suit3) [8] and tertiary protein structure prediction (SPOT-Contact) [9]. However, a lot of effort and resources have been spent on preserving knowledge about proteins by the UniProt Consortium and host institutes (EMBL-EBI, SIB and PIR) through the development of UniProt [10]. Most of this research does not aim at providing information regarding behavioral similarities, such as a specific property similarity. Different approaches have been used for this purpose, including graph machine learning (GraphSol) [11]. Others only researched these properties on E.coli (eSOL) [12].

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Unfortunately, eSOL only aims at predicting E.coli bacteria, while graphSol utilizes state-of-the-art feature extraction techniques such as PSI-BLAST, SPIDER3, and others mentioned earlier to find similar proteins. This process is an extremely time and resource consuming approach, though seemingly performing well.

This paper will therefore not aim to discover alternative proteins when giving a harmful protein. But instead it will discuss a core aspect of it, which is protein property predictions (more specifically the prediction of protein solubility). The methods used in this study can be extended for any other protein properties as well. By inspecting protein solubility prediction, one can explore prediction for other protein features as well.

This raises the following research questions:

1) How to retrieve data for solubility prediction of proteins?
2) How to use protein properties for feature selection?
3) How to improve the protein prediction accuracy?

This paper will explain related works in more detail (chapter II), the methodologies (chapter III), the implementation (chapter IV), experiments (chapter V), a discussion of the results including future work (chapter VII) and a conclusion of the research (chapter IX).

II. Related work

One of the studies that tackles the complex task of predicting solubility of proteins by extracting its various features, is GraphSol [13]. There are different approaches, focusing on the theoretical 3D structure of the protein, or taking evolutionary information by comparing the amino acid (AA) sequence to others and thus inferring the differences and similarities between them. The list of the different methods is almost eternal. However, by combining several of these rigorous methods, different pieces of information of the same protein can be obtained. As a result, solubility prediction from all these features would become feasible.

GraphSol [13] studies how to predict protein solubility by embedding the 3D structure of amino acids. To accomplish this task, the amino acids of the proteins were gathered in a dataset, and features were extracted from the amino acids through several steps. These steps are namely:

1) PSI BLAST (Position-Specific-Iterated BLAST): A scoring matrix obtained from a multiple alignment of the highest scoring hits in an initial BLAST search.

2) HH-Suite3: Obtains a sensitive protein sequence searching based on the pairwise alignment of hidden Markov models (HMMs)
3) SPIDER3: Predicts the local structure of a protein. Prediction on secondary structure and other structural properties of proteins by long short-term memory (LSTM).
4) DCA (Direct Coupling Analysis): Provides a correlated mutation analysis in an AA and codon level. Inducing evolutionary information about the protein.
5) CCMPred (protein residue-residue Contacts from Correlated Mutations Prediction): Markov Random Field pseudo-likelihood maximization for learning protein residue-residue contacts.
6) SPOT-Contact: Provides an accurate prediction of protein contact maps by coupling residual two-dimensional bidirectional LSTM with convolutional neural networks (CNNs).

Once SPOT-Contact is used to calculate the contact map, the map information of proteins is extracted. This information is used in graph convectional networks, where the protein topology attribute graph is constructed. The nodes in the graph represent the structural features which are derived from SPIDER3. These features are formed into a matrix representing the length of sequences and their dimensions. The edges on the other hand, were formed from the outcome of SPOT-Contact. SPOT-Contact outputs whether there is a possibility to make contact between residue pairs in a protein. Therefore, it produces the edges of the graphs. The result is a final graph which shows the prediction from proteins' structures. They are fully connected with the edges' values representing the contact probability of corresponding residue pairs.

DeepSol [14] also refers to the predicted structural properties, being highly related to solubility. The constructed graph from GraphSol [13] is the result of nodes (produced by predicted structural features from SPIDER3) and the edges (produced by using SPOT-Contact to predict contact maps) that can best be used to predict solubility of proteins.

III. Methodology

A. Data

Data utilsed in this paper regarding protein structures and alignments is obtained from the eSOL dataset [15]. Since the goal is finding the solubility of proteins, all the data points with missing solubility values were deleted. The data gathered has 34 features. Including solubility, 23 of these features had no missing values. From these 23, 8 are either unique values like names or descriptions so they were removed because they are not useful for training purposes. From the 15 features left, 13 had numerical value. The usable dataset had 3,149 row records. For training a sequential neural network (SNN), only numerical values can be used. The other two features were inverted to numerical values. To enhance the data already obtained, for every row entry, six extra datapoints were calculated using 7 physicochemical properties of amino acids. [16]

B. Models

The following machine learning approaches were used for making models to predict protein solubility.

- Sequential neural network
- Decision tree
- Random forest
- Linear regression

Because of the restricted data, the models could not be fully trained to accurately (to the extent originally anticipated) predict the solubility of a specific protein. For this matter, multiple models were made and ensemble learning was used to combine the result of these models to get a more accurate result.

To calculate the accuracy of each model, root-mean-square error/deviation (RMSE) and mean absolute error (MAE) were used, this is because finding the solubility of a protein is a regression problem. MSE and RMSE measure the differences between values predicted by a model and the values observed, using the formula introduced below.

$$MAE = \frac{\sum_{i=1}^{N} |x_i - \hat{x_i}|}{N} \tag{1}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \hat{x_i})^2}{N}} \tag{2}$$

where
i = variable i
N = number of data points
x_i = actual observed value
$\hat{x}_i$ = predicted value
The smaller the RMSE and MAE value, the more accurate the model is defined.

1) Sequential Neural Network: Sequential neural networks are artificial neural networks (ANN) that are machine learning algorithms. ANNs use layers of nodes connected by weighted edges. The first layer is the input data, followed by hidden layers and the last layer is the output layer. For purpose of this project, an ANN was implemented in a supervised manner, which is used for prediction or classification tasks when the data available is labeled.

2) Decision Tree: Decision trees are a type of supervised machine learning, where the data is continuously split according to parameters. The leaves of the tree are the final outcomes, or decisions, and the nodes are the parameter being split. There are two types of decision trees, classification trees, and regression trees. For the purpose of this project, the regression tree was used where the outcome variable is continuous.

3) Random Forest: Random forest is a tree based algorithm that combines multiple decision trees to make a decision. As the name also suggests *forest* refers to the forest of trees. Moreover, as the term *random* suggests, these trees are created randomly. The advantage of using multiple random decision trees – or in other words random forest – over a single decision tree is that, random forest can resolve the problem of over-fitting that a single decision tree might cause. Moreover,

random forest is also a very robust method, and it is fast in comparison to other regression models.

4) Linear Regression: Linear regression is a model that calculates the similarities of the features in the first layer of a given neural network, and attempts to predict labels of a different protein by estimating the coefficients of a linear equation. The protein features are represented as data points, and the linear equation defines the relation between properties of the proteins. The equation's independent variables are used to predict the values of the dependent variables. Subsequently, the error is minimized by calculating the RMSE between the two data points.

5) Ensemble Learning: Ensemble learning is used in combination with machine learning models to enhance the output of these models. For this paper, all the models are used in a weighted averaging function to give weights to the prediction of the models. Using the RMSE, the weights are calculated for the ensemble model, where a lower RMSE corresponds to a higher weight in the ensemble model. The prediction on solubility depends on all the models that are assembled in the ensemble model.

IV. Implementation

One of the models implemented is a decision tree (DT). The implementation was done in python using sklearn. This method breaks down a dataset into smaller and smaller subsets, while at the same time an associated decision tree is incrementally developed. The final result is a tree with leaf nodes being decision nodes. To avoid overfitting, a mean sample leaf of 15 was implemented. This way the decision node should have at least 15 data points inside. This number was found after some experiments.

Random forest selects a random subset of the features to use for splitting the data at each node of the tree. This helps to reduce the correlation between the trees and make the ensemble more robust. The algorithm used is also from sklearn. In the latter, the subset of features should have more weight than others cannot be predefined but the maximum number of created "trees" as their maximum depth and the minimum number of data points per tree can be defined.

The next model implemented is sequential neural network (SNN), this model was implemented in python using Tensorflow. SNN was implemented using the same 10 input features as input layer, two hidden layers with each having 64 nodes and 1 output layer which is the solubility value. For the activation function, the rectified linear activation function (ReLU) is used. ReLU is a piecewise linear function that will output the input directly if it is positive, otherwise, it will output zero. This is an appropriate choice because there is no negative prediction. ReLU is the best option for regression problems.

For ensemble learning, each model was given a weight. These weights were calculated using $1 - RMSE$ where the RMSE was normalised. This way the model with lower RMSE has a bigger weight. Then given a protein, the prediction of each model is calculated and multiplied by the weight of its model. The average of all these predictions is calculated for final result. This way the model with smaller RMSE has more influence on final result.

V. Experiments

In order to answer our third research question, the goal of these experiments is to tune the hyper-parameters of the examined algorithms and find their optimal configuration given the dataset.

For the experiments, the data was divided into 5 folds, and a cross validation between the different configurations of the models is performed. This approach was used because it assures that the overall performance does not depend on the split of data used. Furthermore, this approach is widely used in seen literature and is time-reducing as five different configurations are tested at once. In addition, the same seed was used for the models, this means the splitting of the data always happens with same seed of randomness. This make the models' accuracy comparable.

Firstly, for decision trees, the minimum sample leaf was experimented. The minimum sample leaf refers to the minimum amount of data points before nodes can be considered as leaves. In order words, the minimum depth for any branch of the tree. Not much of literature was found to have a better estimate, hence, the threshold was experimented in a range from 0 to 500. The minimum sample split was also tested. It refers to the minimum amount of data nodes before a split of an internal node can occur. To the same extent as with the minimum sample leaf, little literature was found and thus a range of 0 to 450 was tested. In addition, the maximum depth for the decision tree was tested in a range of 0 to 200.

The next set of experiments is also tuning the hyper-paramaters of the RandomForestRegressor algorithm. The parameters tested were the following; the maximum amount of trees in the forest (n estimators), maximum depth of every tree (max depth), min sample leaf, the same as for decision trees, and whether bootstrap samples can be used to build trees or whether to use out-of-bag samples to estimate the generalization score (oobscore). Not much literature was found to have a good first estimation, only in unrelated fields such as weather prediction. It was decided to test every hyper-parameter in a wide range. Given that the cross validation takes 5 different instances of the predictor in one turn, this saved a lot of time while performing experiments. It was also tempted to answer the second research question, how to use the protein properties for feature selection. Nevertheless, because of the implementation used, this is not possible, since the subset of features used for the splitting of branches is automatic in the sklearn package, it is implementation specific and could not be changed.

Another method, used to test the accuracy between two feature parameters, was linear regression. The RMSE was calculated by testing different batches, whilst keeping the epochs the same and learning rate at 0.01 (the parameters can be changed). Since the linear regression only contains one layer from the sequential keras neural network, and no

activation function, the result is expected to be the least accurate of all models. This large uncertainty is due to the fact that this model attempts to implement a line of best fit between protein features. However, because the protein features themselves contain various complexity, the expected accuracy level is low.

VI. Results

The data for the graphs below can be found in the GitHub link provided in chapter X in the directory ExperimentsEatingInsects.
For figure 1 to figure 5 the negative value of RMSE is displayed, hence 0 means no error, and the bigger the value better the results. For the last figure, figure 6 the value of RMSE and MAE are displayed.

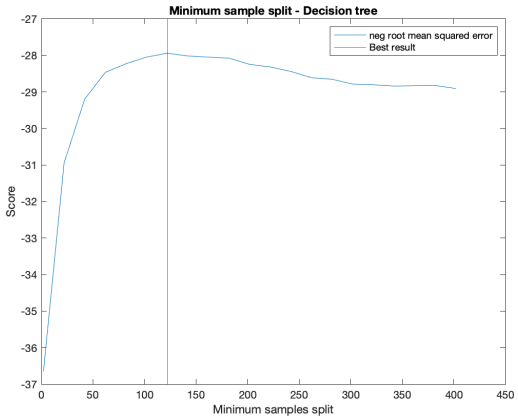


Fig. 1: Testing for the best minimum sample split in decision trees: best result 122

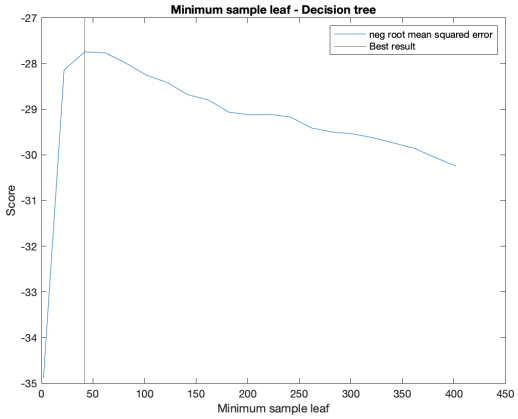


Fig. 2: Testing for the best minimum sample leaf in decision trees: best result 42

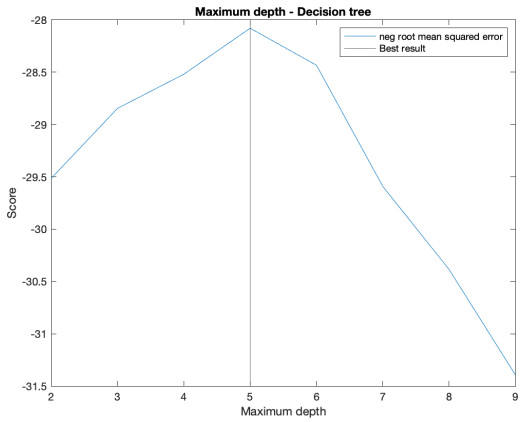


Fig. 3: Testing for the maximum depth in random forest: best result 5

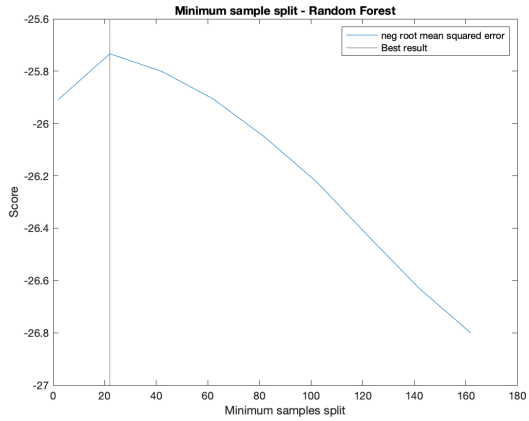


Fig. 4: Testing for the minimum sample split in random forest: best result 22

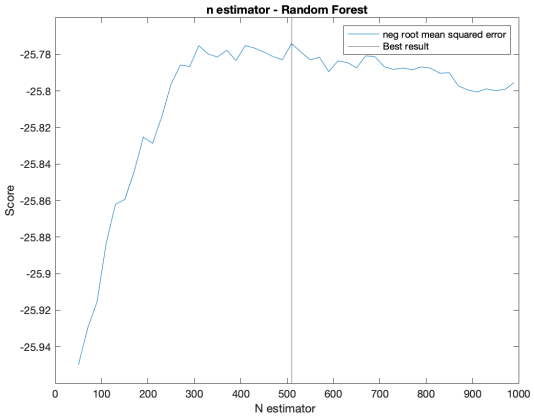


Fig. 5: Testing for the maximum number of trees in random forest: best result 510

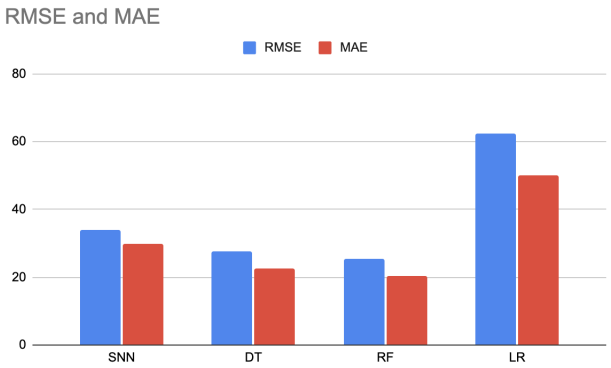


Fig. 6: Root mean square error (RMSE) and mean absolute error (MAE) of all the models implemented. The exact values can be found in appendix A.

VII. Discussion

The best estimator for the decision trees, was the one with the minimum sample split set to 122. When looking at the first graph from the results, it can be seen that there is a steady performance in the range from 100 to 200. While testing for minimum sample leaf, the minimum sample split was set to 122, the best predictor had a threshold value of 42. Subsequently, with these two parameters determined, the maximum depth found optimal was 5. The corresponding the graph in results only shows the interval between 0 and 10, because after 5 it was a continuous decrease in performance. This might be due to overfitting, if there is no restriction in depth of the decision tree it overfits all possible data points.

From the other set, the minimum sample split it was found to be optimal at 22, the performance did not oscillate too much with the overall test, a range between 15-30 was found to be the most appropriate. From the iterations performed, the minimum sample leaf did not vary much once it was bigger than 40. Subsequently, the maximum amount of trees per forest was also not found to significantly alter the performance of the random forest, although, the best was found to be bigger than 300. These latter graphs were not included in the report since they showed a linear or constant behaviour. Furthermore, it was concluded after several repetitions that the out-of-bag score was not helping the prediction nor making it worse, so it was set to false, as the default of the sklearn package. On the other side, the bootstrap did aid the performance, thus it was kept true.

For the linear regression model, the $RMSE = 62.241772$ and the mean absolute error was $MAE = 50.064236$. These results with a large uncertainty were expected, because the linear model for the protein features is the least accurate for datasets with small linear dependencies. Therefore, the model computation is optimal for calculations when working with a massive dataset to decrease time complexity. Correspondingly, linear regression has an advantage in training data in short time intervals. However, the main disadvantage of this model is that it is highly inaccurate at predicting labels.

VIII. Limitations and Future works

One of points that was not considered by this study, was recognising the property similarities between the proteins. The structure of proteins contribute a monumental role in predicting their properties and vice versa. Therefore a protein's properties are indicators of their structures and likewise their structure can determine their properties. As for the future works, one of the methods that can be applied, in order to predict protein solubility, is *unsupervised graph machine learning*. Each protein (with its entire amino acid sequence) would be considered as a node in the graph. Then the method continues by clustering proteins into groups based on how similar their properties are to each other. For each property, a limit range would be set to measure and compare how similar they are to each other. As an instance, a limit can be set for solubility so that once two (or more) proteins are within the range of solubility limit, they will be considered as similar to each other. Furthermore, using unsupervised machine learning methods (such as k-NN) nodes in the graph will be clustered together. The clustering is based on how many properties each group of nodes have in common to each other. An edge would be established between similar nodes with a value that indicates how many properties they have in common. Once the graph is built up, the solubility (or any other properties in question) of an unknown protein could be predicted by inspecting which cluster they fit into.

Another method that can be explored further is using the algorithms that *natural language processing* (NLP) provides for solubility prediction. As it was explained in chapter II, the 3D structure of proteins can be analysed through multiple steps. These steps take the amino acid sequence to predict the primary and secondary structures of proteins. Then by using the contact map, the tertiary structure (which represents the protein features as well) would be extracted. The result will be a graph with multiple nodes (each representing the predicted amino-acid sequence) and their alignments (which represents their 3D structures). Once these features for every protein sequence in the database is extracted, an encoding function can be called to vectorize the sequences and their alignments for each graph. Afterwards, machine learning algorithms can be trained on the vectors and their solubility measures. The solubility of unknown protein sequences then can be predicted by their structural resemblance to the trained dataset.

The models that were introduced in this study had several limitations, which affected the accuracy and precision of the observations. The first limitation was that the dataset contained very limited records. Thus, the data was not sufficient enough to draw precise predictions. To combat this challenge a separate dataset had to be extrapolated (i.e the eSOL dataset). Furthermore, the dataset used was specifically catered towards the study and research of E.coli (a type of bacteria common in the environment) based proteins. Hence, there is a bias present throughout the observations, such that all the similarity features are oriented with respect to the E.coli proteins within the eSOL dataset. Consequently, there may be some data over-

fitting present in obtained results, and the trained data on the respective models. However, the models can be utilised on other datasets by simply using other datasets as an input and train data for the models.

The ensemble learning method that was presented in this study, is a simple version that combines multiple algorithms together, in order to make predictions. This model can be improved significantly by just adding other statistical methods (such as Bayesian or support-vector-regression). As it was also explained in chapter IV that the model uses a simple linear strategy for combining multiple algorithms. This can be improved by using more statistical methods and combining them using an boosting strategy (which involves adding models sequentially and using correction on the outputs of prior models). The boosting strategy weights the models' outputs and averages them in order to make the final predictions. Other strategies that can be explored are namely:

- bagging [17]
- robust semi-supervised SVM [18]
- AdaBoost and random forest [19]

IX. Conclusion

In this paper, we proposed a protein property prediction tool that gives a solubility prediction from a given protein input. Retrieving this data is a time-consuming and resource-heavy task to perform, with almost no valuable data that could be used to work with. For this paper, various sources were used to retrieve several thousands of data samples.

Once the data is retrieved, its protein and amino acid properties can be utilized as features for assumed protein property predictions, while giving individual models the opportunity to attempt and find correlations with protein property values and solubility properties.

Unfortunately, the performances of each individual model are of poor quality due to the lack of significantly useful data. One promising way of improving this accuracy, or at least providing a standardized prediction is through the use of ensemble learning.

X. Data and Material

The source code and data for experiments are available at https://github.com/nscharrenberg/EatingInsects.

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APPENDIX A
EXPERIMENTS

A. RMSE & MAE table

	RMSE	MAE
Sequential neural network	34.02	29.86
Decision Tree	27.68	22.65
Random forest	25.29	20.52
Linear regression	62.24	50.06